Nishant Gadde

Student Researcher — Continual Learning, Representation Learning and Deep Network Theory

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Education

The University of Texas at Austin Expected May 2028

B.S. Electrical and Computer Engineering (Engineering Honors) and Mathematics

- GPA 3.87/4.0. Coursework: Probability & Random Processes, Optimization, Linear Algebra, Complex Analysis, Numerical Methods, Reinforcement Learning.
- USACO Platinum Division (top 1%), ranked top 100 worldwide, USA Computing Olympiad, 2024–2025; foundations in search, planning, and agent decision-making.

Experience

Research Assistant, Vision–Language Agents Jan 2026 – Present

Oden Institute for Computational Engineering & Sciences, UT Austin, Austin, TX

- Co-developed affordance-centric representations for vision–language agents, improving success on multi-step tool-use tasks.
- Designed uncertainty-calibrated routing that selects between fast and slow agents, achieving near-oracle accuracy at lower latency.
- Contributed to open-source agent harnesses for interactive visual reasoning, including memory and context compression.

Machine Learning Engineer Intern, AI Infrastructure Jun 2026 – Aug 2026

Apple, Seattle, WA

- Formulated batching and caching as a scheduling optimization over LLM inference workloads, raising GPU utilization.
- Characterized throughput–latency tradeoffs for multi-tenant agentic services, sustaining 99.99% reliability.

Machine Learning Engineer Intern Jun 2025 – Aug 2025

Rayfield Systems, Houston, TX

- Built recurrent state-space models for high-frequency stochastic processes, beating regression baselines and informing RL.

Research Assistant, Sustainable Power Systems Jun 2024 – Aug 2025

Stanford Sustainability Systems Laboratory (S3L), Remote

- Co-authored IEEE CAI work on ML approximations of AC optimal power flow with <1% optimality gaps and 1000× faster solves.

Selected Publications

- Siva, R.*, Bhatt, N. P.*, Yang, Y., Lee, S., **Gadde, N.**, Ellis, C., Velasquez, A., Wang, Z., & Topcu, U. "What Objects Enable, Not What They Are: Functional Latent Spaces for Affordance Reasoning," arXiv, 2026.
- Miltner, M., Vasata, D., Friedjungová, M., **Gadde, N.**, Stogl, O., Starý, O., & Rajagopal, R. "Towards Real-Time Machine Learning Approximations of AC Optimal Power Flow," IEEE CAI, 2025.

Selected Projects

Hamiltonian Flow Matching for Sustainable Grid Control 2026

PyTorch, RL, generative modeling

- Proposed Hamiltonian Flow Matching (HFM), an energy-regularized flow-matching objective enforcing Kirchhoff consistency.
- Designed an adaptive curvature-based inference scheduler that minimizes GPU energy per scenario while preserving forecast accuracy.
- Built the GridControl-RL benchmark coupling scenario generation with SAC-based dispatch agents to evaluate accuracy.

Claude-Orchestrated Multi-Agent Research Assistant 2025–2026

Python, REST, tool-use, memory

- Engineered an agent harness where planner, researcher, and editor agents coordinate via shared memory and message-passing.
- Implemented context compression, citation tracking, and task decomposition policies; reduced manual supervision.
- Ran quantitative evals on coding and literature-review benchmarks, comparing harness variants and prompt strategies.

Technical Skills

Machine Learning: PyTorch, TensorFlow, RL (SAC, PPO), sequence modeling, generative modeling, uncertainty quantification, representation learning.

ML & Systems: PyTorch, TensorFlow, XGBoost, Spark, Kafka, Linux, Git, CMake, NumPy, pandas, SciPy, scikit-learn.

Programming: Python, C++, SQL, Rust, Java, MATLAB, Bash.

Systems: Python, C++, Rust, SQL, Linux, CUDA profiling, distributed inference, Kafka, Spark, Git, CMake.